

Functions & Graphs

ANSWER KEY



Evaluating functions

- 1 If $f(x) = 3x - 2$, find $f(4)$.

$$f(4) = 3(4) - 2 = 10.$$

- 2 If $f(x) = x^2 + 2x$, find $f(-3)$.

$$f(-3) = 9 - 6 = 3.$$

- 3 Consider $f(x) = 2x^2 - 5x + 1$. Evaluate:

a $f(0)$ 1

b $f(2)$ -1

Domain and range

- 4 The graph of $y = \sqrt{x + 3}$ is shown below. State the domain and range based on the graph, then verify the domain algebraically.

From the graph, the domain is $x \geq -3$ and the range is $y \geq 0$. Algebraically, the expression under the root must be non-negative: $x + 3 \geq 0$, so $x \geq -3$, confirming the graph.

- 5 Find the domain of $f(x) = \frac{1}{x - 5}$.

The denominator cannot equal zero, so $x \neq 5$. Domain: all real numbers except 5.

- 6 Find the domain of $f(x) = \sqrt{2x - 8}$.

$$2x - 8 \geq 0, \text{ so } x \geq 4.$$

Composition of functions

- 7 If $f(x) = 2x + 1$ and $g(x) = x^2 - 3$, find $(f \circ g)(2)$.

$$g(2) = 4 - 3 = 1. f(1) = 2(1) + 1 = 3.$$

- 8 If $f(x) = x + 5$ and $g(x) = 3x$, find $(g \circ f)(x)$ as a simplified expression.

$$g(f(x)) = 3(x + 5) = 3x + 15.$$

- 9 If $f(x) = x^2$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and $(g \circ f)(x)$, and state whether they are equal.

$f(g(x)) = (x - 4)^2 = x^2 - 8x + 16$. $g(f(x)) = x^2 - 4$. These are not equal, so function composition is generally not commutative.

Inverse functions

- 10 Find the inverse of $f(x) = 4x - 8$.

Swap x and y in $y = 4x - 8$: $x = 4y - 8$, so $y = \frac{x+8}{4}$. $f^{-1}(x) = \frac{x+8}{4}$.

- 11 Find the inverse of $f(x) = \frac{x-3}{2}$.

Swap: $x = \frac{y-3}{2}$, so $2x = y - 3$, giving $y = 2x + 3$. $f^{-1}(x) = 2x + 3$.

- 12 Verify that $f(x) = 5x + 1$ and $g(x) = \frac{x-1}{5}$ are inverses of each other by computing $f(g(x))$.

$f(g(x)) = 5\left(\frac{x-1}{5}\right) + 1 = (x-1) + 1 = x$, confirming they are inverses.

Transformations of graphs

- 13 The graph of $y = f(x)$ is shifted 4 units left and 2 units down. Write the new function in terms of f .

$$y = f(x+4) - 2.$$

- 14 If $f(x) = x^2$ and $g(x) = -f(x) + 3$, describe the transformation from f to g .

g reflects f over the x -axis, then shifts the result up 3 units.

Piecewise functions

- 15 A function is defined by $f(x) = x + 2$ for $x < 0$ and $f(x) = x^2$ for $x \geq 0$. Find $f(-3)$ and $f(3)$.

$f(-3) = -3 + 2 = -1$ (using the first piece, since $-3 < 0$). $f(3) = 3^2 = 9$ (using the second piece, since $3 \geq 0$).

- 16 A function is defined by $f(x) = 2x$ for $x \leq 2$ and $f(x) = x + 5$ for $x > 2$. Find $f(2)$ and $f(5)$, and state whether f is continuous at $x = 2$.

$f(2) = 2(2) = 4$ (using the first piece, since $2 \leq 2$). $f(5) = 5 + 5 = 10$. As $x \rightarrow 2$ from the right, $f(x) \rightarrow 2 + 5 = 7$, which does not match $f(2) = 4$, so f is not continuous at $x = 2$ — there is a jump.

Reading function graphs

- 17 The graph of $f(x) = x^2 - 4$ is shown below. Use the graph to estimate the x -intercepts and y -intercept, then find them exactly.

x -intercepts: solve $x^2 - 4 = 0$, giving $x = 2$ and $x = -2$. y -intercept: $f(0) = -4$.

Even, odd, and other function properties

- 18 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither, using $f(-x)$.

$$f(-x) = (-x)^2 - 3 = x^2 - 3 = f(x), \text{ so } f \text{ is even.}$$

- 19 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.

$$f(-x) = -x^3 + x = -(x^3 - x) = -f(x), \text{ so } f \text{ is odd.}$$

A well-designed word problem: modeling a car rental cost

- 20 This question has three parts. A rental car company models the cost of a rental as $C(d) = 45d + 20$ dollars, where d is the number of days rented. (a) Find $C(5)$, and interpret its meaning in context. (b) Define $P(d) = C(d) - 200$, representing profit relative to a fixed cost of 200 dollars per rental period. Write $P(d)$ explicitly and find $P(5)$. (c) Find the inverse function $C^{-1}(x)$, and use it to determine how many days someone rented the car if they were charged 335 dollars.

(a) $C(5) = 45(5) + 20 = 245$. Renting the car for 5 days costs 245 dollars total. (b)

$P(d) = 45d + 20 - 200 = 45d - 180$. $P(5) = 45(5) - 180 = 225 - 180 = 45$ dollars. (c) From $x = 45d + 20$, solve for d : $d = \frac{x - 20}{45}$, so $C^{-1}(x) = \frac{x - 20}{45}$. For $x = 335$: $d = \frac{335 - 20}{45} = \frac{315}{45} = 7$ days.

Multiple choice — easy

- 21 If $f(x) = -4x^2 + 3x - 4$, find $f(7)$.

a. -177

b. -179 ✓

c. -181

d. -11

$$f(7) = -4(7)^2 + 21 - 4 = -179.$$

22 If $f(x) = x^2 + 3x - 4$, find $f(-1)$.

- a. -8
- b. -3
- c. -6 ✓
- d. -4

$$f(-1) = (-1)^2 - 3 - 4 = -6.$$

23 If $f(x) = 3x^2 - 4x + 5$, find $f(6)$.

- a. 87
- b. 89 ✓
- c. 91
- d. -1

$$f(6) = 3(6)^2 - 24 + 5 = 89.$$

24 If $f(x) = -2x^2 + 5x - 2$, find $f(-3)$.

- a. -11
- b. -35 ✓
- c. -33
- d. -37

$$f(-3) = -2(-3)^2 - 15 - 2 = -35.$$

25 If $f(x) = x^2 - 9x$, find $f(8)$.

- a. -64
- b. -6
- c. -8 ✓
- d. -10

$$f(8) = (8)^2 - 72 = -8.$$

26 If $f(x) = -4x^2 - 5x - 6$, find $f(2)$.

a. -34

b. -30

c. -24

d. -32 ✓

$$f(2) = -4(2)^2 - 10 - 6 = -32.$$

27 If $f(x) = -4x^2 + 8x - 8$, find $f(4)$.

a. -42

b. -38

c. 8

d. -40 ✓

$$f(4) = -4(4)^2 + 32 - 8 = -40.$$

28 If $f(x) = 4x^2 + x - 9$, find $f(1)$.

a. -2

b. -4 ✓

c. -1

d. -6

$$f(1) = 4(1)^2 + 1 - 9 = -4.$$

29 If $f(x) = -3x^2 - 5x + 9$, find $f(-7)$.

a. -105

b. -103 ✓

c. 65

d. -101

$$f(-7) = -3(-7)^2 + 35 + 9 = -103.$$

30 If $f(x) = -4x^2 - 7x + 3$, find $f(8)$.

a. -309 ✓

b. -307

c. -311

d. -85

$$f(8) = -4(8)^2 - 56 + 3 = -309.$$

31 If $f(x) = -x^2 + x$, find $f(-1)$.

a. 0

b. -4

c. 1

d. -2 ✓

$$f(-1) = -(-1)^2 - 1 = -2.$$

32 If $f(x) = -2x^2 - 3x + 1$, find $f(6)$.

a. -29

b. -87

c. -89 ✓

d. -91

$$f(6) = -2(6)^2 - 18 + 1 = -89.$$

33 If $f(x) = -3x^2 - 6x + 1$, find $f(-7)$.

- a. -102
- b. -106
- c. 64
- d. -104 ✓

$$f(-7) = -3(-7)^2 + 42 + 1 = -104.$$

34 If $f(x) = -4x^2 + 3x + 7$, find $f(8)$.

- a. -223
- b. -227
- c. -1
- d. -225 ✓

$$f(8) = -4(8)^2 + 24 + 7 = -225.$$

35 If $f(x) = -4x^2 - 5x + 6$, find $f(-1)$.

- a. 5
- b. 7 ✓
- c. 9
- d. 15

$$f(-1) = -4(-1)^2 + 5 + 6 = 7.$$

Multiple choice — medium

36 If $f(x) = x - 2$ and $g(x) = 4x^2 + 3$, find $(f \circ g)(-6)$.

a. 148

b. 147

c. 145 ✓

d. 143

$$g(-6) = 4(-6)^2 + 3 = 147. f(147) = (147) - 2 = 145.$$

37 If $f(x) = 3x + 1$ and $g(x) = x^2 + 4$, find $(f \circ g)(-1)$.

a. 16 ✓

b. 14

c. 18

d. 5

$$g(-1) = (-1)^2 + 4 = 5. f(5) = 3(5) + 1 = 16.$$

38 If $f(x) = -4x + 5$ and $g(x) = 3x^2 + 6$, find $(f \circ g)(-5)$.

a. -319 ✓

b. -317

c. 81

d. -321

$$g(-5) = 3(-5)^2 + 6 = 81. f(81) = -4(81) + 5 = -319.$$

39 If $f(x) = 4x - 3$ and $g(x) = 2x^2 + 2$, find $(f \circ g)(0)$.

a. 5 ✓

b. 3

c. 7

d. 2

$$g(0) = 2(0)^2 + 2 = 2. f(2) = 4(2) - 3 = 5.$$

40 If $f(x) = 2x - 8$ and $g(x) = x^2 + 5$, find $(f \circ g)(4)$.

a. 21

b. 36

c. 34 ✓

d. 32

$$g(4) = (4)^2 + 5 = 21. f(21) = 2(21) - 8 = 34.$$

41 If $f(x) = -3x - 9$ and $g(x) = -4x^2 + 2$, find $(f \circ g)(-3)$.

a. 91

b. -34

c. 95

d. 93 ✓

$$g(-3) = -4(-3)^2 + 2 = -34. f(-34) = -3(-34) - 9 = 93.$$

42 If $f(x) = -2x - 6$ and $g(x) = 2x^2 + 8$, find $(f \circ g)(-6)$.

a. -164

b. -166 ✓

c. -168

d. 80

$$g(-6) = 2(-6)^2 + 8 = 80. f(80) = -2(80) - 6 = -166.$$

43 If $f(x) = -4x + 3$ and $g(x) = -2x^2 - 5$, find $(f \circ g)(-1)$.

a. 29

b. 31 ✓

c. 33

d. -7

$$g(-1) = -2(-1)^2 - 5 = -7. f(-7) = -4(-7) + 3 = 31.$$

44 If $f(x) = -4x + 2$ and $g(x) = 2x^2 + 1$, find $(f \circ g)(0)$.

a. 1

b. -4

c. 0

d. -2 ✓

$$g(0) = 2(0)^2 + 1 = 1. f(1) = -4(1) + 2 = -2.$$

45 If $f(x) = 2x + 2$ and $g(x) = -3x^2 - 7$, find $(f \circ g)(-1)$.

a. -20

b. -18 ✓

c. -10

d. -16

$$g(-1) = -3(-1)^2 - 7 = -10. f(-10) = 2(-10) + 2 = -18.$$

46 If $f(x) = -x + 3$ and $g(x) = 2x^2 + 3$, find $(f \circ g)(-2)$.

a. -8 ✓

b. -10

c. 11

d. -6

$$g(-2) = 2(-2)^2 + 3 = 11. f(11) = -(11) + 3 = -8.$$

47 If $f(x) = 2x + 2$ and $g(x) = x^2 + 4$, find $(f \circ g)(-3)$.

a. 26

b. 30

c. 13

d. 28 ✓

$$g(-3) = (-3)^2 + 4 = 13. f(13) = 2(13) + 2 = 28.$$

48 If $f(x) = 3x - 4$ and $g(x) = -2x^2 - 2$, find $(f \circ g)(1)$.

a. -16 ✓

b. -18

c. -4

d. -14

$$g(1) = -2(1)^2 - 2 = -4. f(-4) = 3(-4) - 4 = -16.$$

49 If $f(x) = -x + 9$ and $g(x) = 2x^2 + 7$, find $(f \circ g)(-4)$.

a. -32

b. -30 ✓

c. 39

d. -28

$$g(-4) = 2(-4)^2 + 7 = 39. f(39) = -(39) + 9 = -30.$$

50 If $f(x) = -x - 3$ and $g(x) = 3x^2$, find $(f \circ g)(0)$.

- a. -1
- b. -5
- c. 0
- d. -3 ✓

$$g(0) = 3(0)^2 = 0. f(0) = -(0) - 3 = -3.$$

Multiple choice — hard

51 Find the inverse of $f(x) = 3x - 7$.

- a. $\frac{x+7}{-3}$
- b. $\frac{x+7}{3}$ ✓
- c. $\frac{x-7}{3}$
- d. $3x - 7$

$$\text{Swap } x \text{ and } y \text{ in } y = 3x - 7: x = 3y - 7, \text{ so } y = \frac{x+7}{3}. f^{-1}(x) = \frac{x+7}{3}.$$

52 Find the inverse of $f(x) = 3x + 4$.

- a. $\frac{x-4}{-3}$
- b. $\frac{x-4}{3}$ ✓
- c. $\frac{x+4}{3}$
- d. $3x + 4$

$$\text{Swap } x \text{ and } y \text{ in } y = 3x + 4: x = 3y + 4, \text{ so } y = \frac{x-4}{3}. f^{-1}(x) = \frac{x-4}{3}.$$

53 Find the inverse of $f(x) = 4x - 2$.

- a. $\frac{x+2}{-4}$
- b. $\frac{x+2}{4}$ ✓
- c. $4x - 2$
- d. $\frac{x-2}{4}$

Swap x and y in $y = 4x - 2$: $x = 4y - 2$, so $y = \frac{x+2}{4}$. $f^{-1}(x) = \frac{x+2}{4}$.

54 Find the inverse of $f(x) = -2x - 5$.

- a. $-2x - 5$
- b. $\frac{x-5}{-2}$
- c. $\frac{x+5}{-2}$ ✓
- d. $\frac{x+5}{2}$

Swap x and y in $y = -2x - 5$: $x = -2y - 5$, so $y = \frac{x+5}{-2}$. $f^{-1}(x) = \frac{x+5}{-2}$.

55 Find the inverse of $f(x) = 4x + 9$.

- a. $\frac{x-9}{-4}$
- b. $\frac{x+9}{4}$
- c. $\frac{x-9}{4}$ ✓
- d. $4x + 9$

Swap x and y in $y = 4x + 9$: $x = 4y + 9$, so $y = \frac{x-9}{4}$. $f^{-1}(x) = \frac{x-9}{4}$.

56 Determine whether $f(x) = 2x^3 + 9x$ is even, odd, or neither.

- a. even
- b. both even and odd
- c. neither
- d. odd ✓

$f(-x) = 2(-x)^3 + 9(-x) = -f(x)$, so f is odd.

57 Determine whether $f(x) = 2x^4 + 7x^2$ is even, odd, or neither.

a. both even and odd

b. neither

c. odd

d. even ✓

$f(-x) = 2(-x)^4 + 7(-x)^2 = f(x)$, so f is even.

58 Determine whether $f(x) = x^2 + 4x$ is even, odd, or neither.

a. even

b. both even and odd

c. neither ✓

d. odd

$f(-x) = (-x)^2 + 4(-x)$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

59 Determine whether $f(x) = -x^3 + 2$ is even, odd, or neither.

a. both even and odd

b. even

c. neither ✓

d. odd

$f(-x) = -(-x)^3 + 2$, which equals neither $f(x)$ nor $-f(x)$, so f is neither even nor odd.

60 Determine whether $f(x) = -2x^2 + 8$ is even, odd, or neither.

a. both even and odd

b. odd

c. even ✓

d. neither

$f(-x) = -2(-x)^2 + 8 = f(x)$, so f is even.

61 A function is defined by $f(x) = 5x + 9$ for $x < 5$ and $f(x) = -3x + 5$ for $x \geq 5$. Find $f(2)$ and $f(5)$.

a. 19, -10 ✓

b. 21, -10

c. -10, 19

d. 19, -12

$f(2) = 5(2) + 9 = 19$ (using the first piece, since $2 < 5$). $f(5) = -3(5) + 5 = -10$ (using the second piece, since $5 \geq 5$).

62 A function is defined by $f(x) = -x$ for $x < -3$ and $f(x) = -5x - 5$ for $x \geq -3$. Find $f(-4)$ and $f(2)$.

a. 6, -15

b. 4, -17

c. 4, -15 ✓

d. -15, 4

$f(-4) = -(-4) = 4$ (using the first piece, since $-4 < -3$). $f(2) = -5(2) - 5 = -15$ (using the second piece, since $2 \geq -3$).

63 A function is defined by $f(x) = x + 3$ for $x < 0$ and $f(x) = x + 1$ for $x \geq 0$. Find $f(-1)$ and $f(0)$.

- a. 1, 2
- b. 2, -1
- c. 2, 1 ✓
- d. 4, 1

$f(-1) = (-1) + 3 = 2$ (using the first piece, since $-1 < 0$). $f(0) = (0) + 1 = 1$ (using the second piece, since $0 \geq 0$).

64 A function is defined by $f(x) = 2x + 1$ for $x < 0$ and $f(x) = -3x - 6$ for $x \geq 0$. Find $f(-5)$ and $f(3)$.

- a. -9, -15 ✓
- b. -9, -17
- c. -7, -15
- d. -15, -9

$f(-5) = 2(-5) + 1 = -9$ (using the first piece, since $-5 < 0$). $f(3) = -3(3) - 6 = -15$ (using the second piece, since $3 \geq 0$).

65 A function is defined by $f(x) = -3x + 7$ for $x < -5$ and $f(x) = 5x - 9$ for $x \geq -5$. Find $f(-9)$ and $f(-2)$.

- a. 36, -19
- b. 34, -19 ✓
- c. -19, 34
- d. 34, -21

$f(-9) = -3(-9) + 7 = 34$ (using the first piece, since $-9 < -5$). $f(-2) = 5(-2) - 9 = -19$ (using the second piece, since $-2 \geq -5$).